CARMA: Using Ground Based Spectroscopy to Bring Positive Change in the World

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INTRODUCTION TO STELLA

STELLA (Science and Technology Education for Land/Life Assessment) is a NASA developed spectrometer that measures a variety of wavelengths on the electromagnetic spectrum that are integral in determining plant health. By utilizing near-infrared and visible red, blue, and green wavelengths, STELLA can determine a plant's level of stress before there are visible signs. These near infrared and visible light spectra are indicative of chlorophyll absorption in a plant.

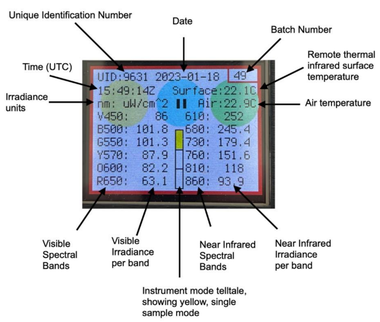

Figure 1: STELLA Display

Figure 2: STELLA 1.1 Device

CARMA MISSION

Our goal, as the CARMA (Climate Adaptation and Resilience Monitoring Alliance) team, has been to validate the STELLA device as a research-grade instrument. To do this, we are conducting field studies in areas of interest, doing accuracy testing, data collection and analysis.

Additionally, STELLA has been tested against a spectral radiometer to determine its variance from those known values. CARMA aims to better equip those working in agriculture with tools that can grant them earlier detection of plant health threats, allowing them to better protect their plants against the threats of climate change.

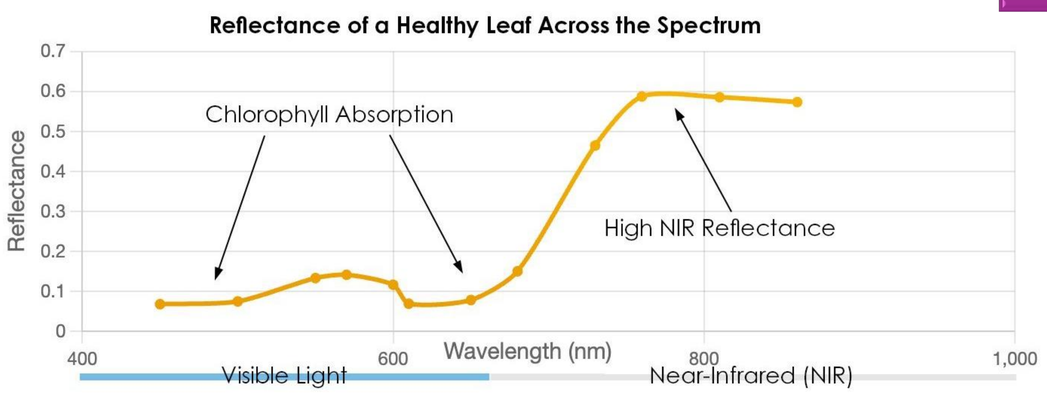

Figure 3: Healthy Leaf Reflectance Diagram

STELLA IMPACT

The STELLA device is a good way to introduce anyone to spectroscopy. If approved for research, then people can build and use their own STELLA devices to help accumulate data on plant health, urban heat island effect, wildfire intensity, and other environmental applications within their own communities.

Our team is utilizing STELLA to provide point-based data on specific plants in areas of interest, which can be used in conjunction with Landsat to provide higher spatial resolution, encapsulating the health of an area more completely.

CARMA RESEARCH PLAN

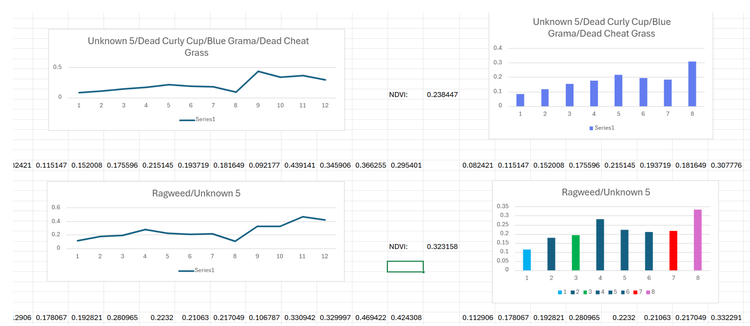

Figure 4: Two Examples of Reflectance Data Collected by CARMA

STELLA's specific scope means that multiple plants can be captured within the same measurement, so all species captured are recorded and considered in the analysis of the collected data.

CARMA will be conducting future research in the following manner:

Research Plan

Greenhouse comparative plant study	Field data of overgrazing impact	Field data of drought impact

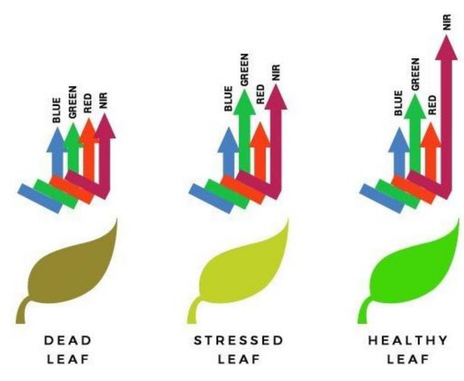

Figure 5: Healthy Leaf Reflectance Diagram

AGRICULTURAL APPLICATION

Ranches such as our stakeholder, Rafter W. Ranch in Simla, CO, are dedicated to leaving overgrazing behind, instead using alternative grazing patterns for cattle to avoid this. The STELLA device is a useful tool which, through repeated measurements of the health of grass over time, allows ranchers and farmers to tell if their mitigation of overgrazing is working.

We are working with our stakeholder to record the new growth of native plants and note the new wildlife that appears on the ranch. The owner, Lance Wheeler, is dedicated to the preservation of Colorado's biodiversity in both flora and fauna. Farms like these mean that we must provide new resources as well, which is why we are developing a website for easier ways to process data. Ranchers and farmers will be able to access their STELLA data, upload it to the website, and see the health of their plants.

Figure 6: CARMA Field Testing

Figure 7: Lance Wheeler

FUTURE WORK

Learn more about CARMA here!

REFERENCES

References:
Figure 1: NASA
Figure 2: NASA
Figure 3: Selkirk Geospatial Research Center
https://www.sgrc.selkirk.ca/blog/object-classification-using-spectral-signatures/
Figure 5: USGS, NASA